# Class 8

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```r
getwd()
```

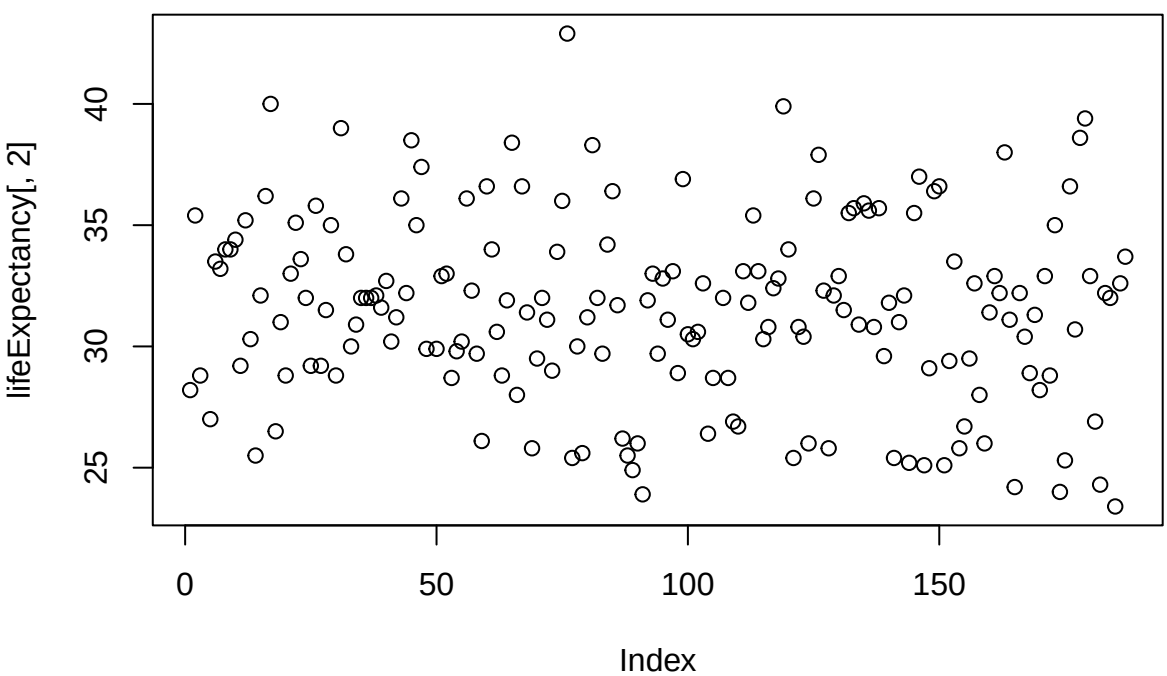
```
## [1] "/home/dhruvi/Desktop/Biostatistics/sessions"
```

```r
lifeExpectancy <- read.csv("../GapMinder/life_expectancy_years.csv", header = TRUE, check.names = FALSE
```

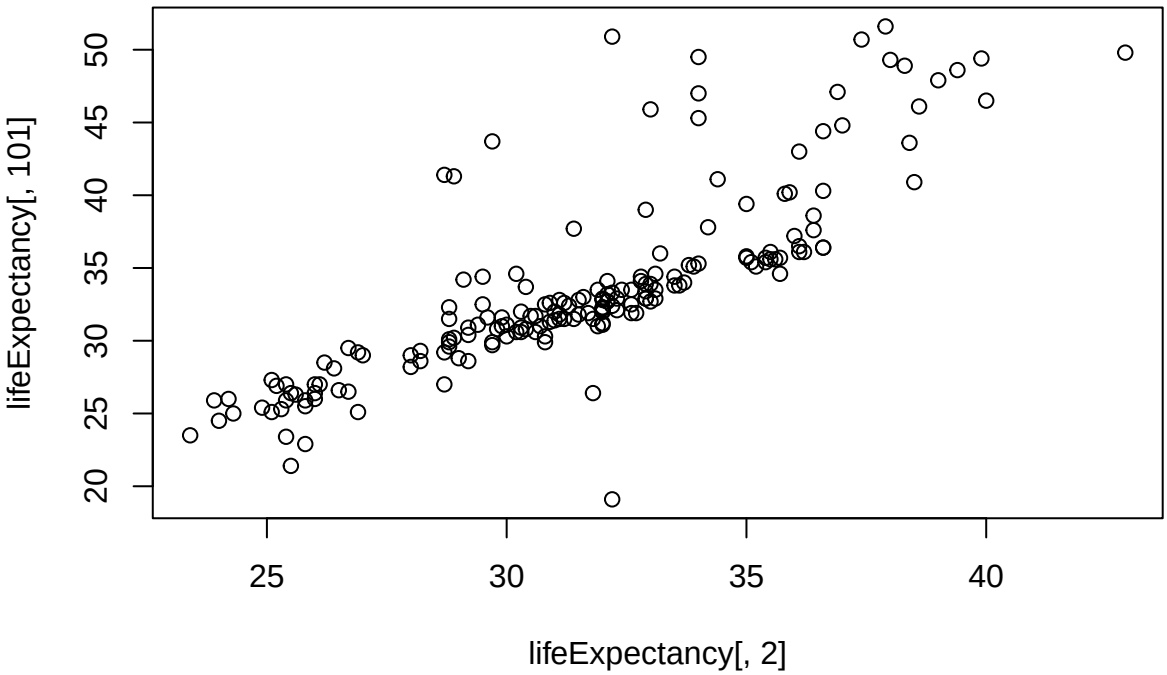
```r
head(lifeExpectancy[lifeExpectancy$country == "Sweden"])
```

```
##   1960
## 1 39.3
## 2 62.2
## 3 52.5
## 4   NA
## 5 40.6
## 6 63.3
```

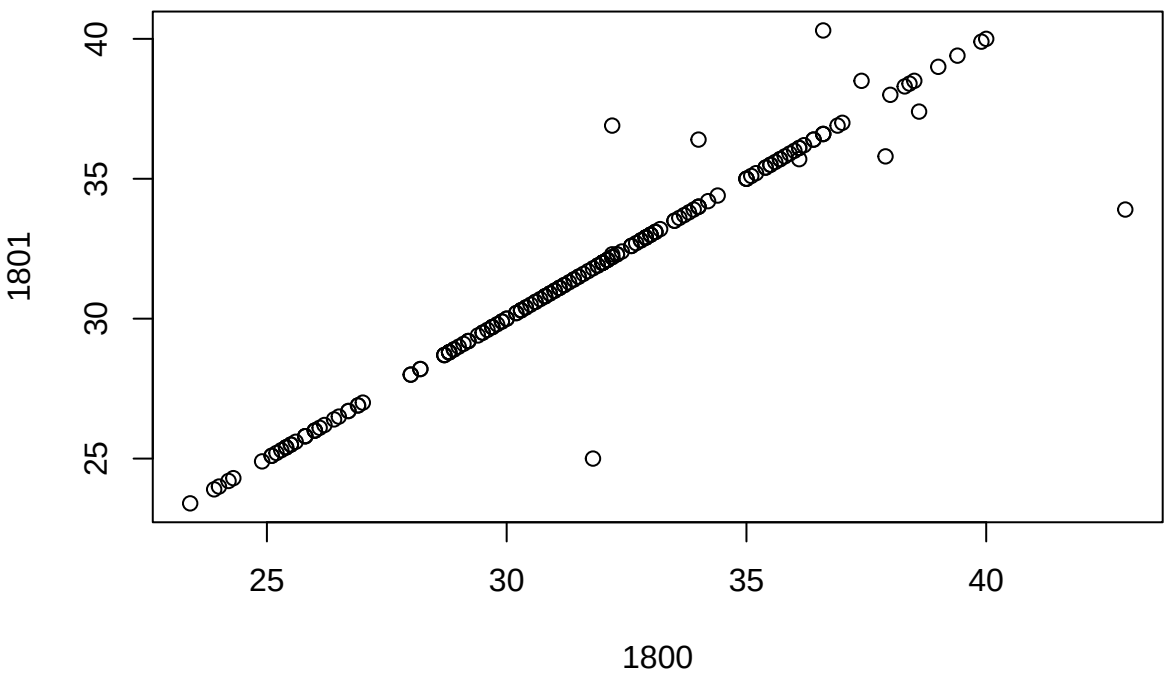
```r
plot(lifeExpectancy[,2])
```



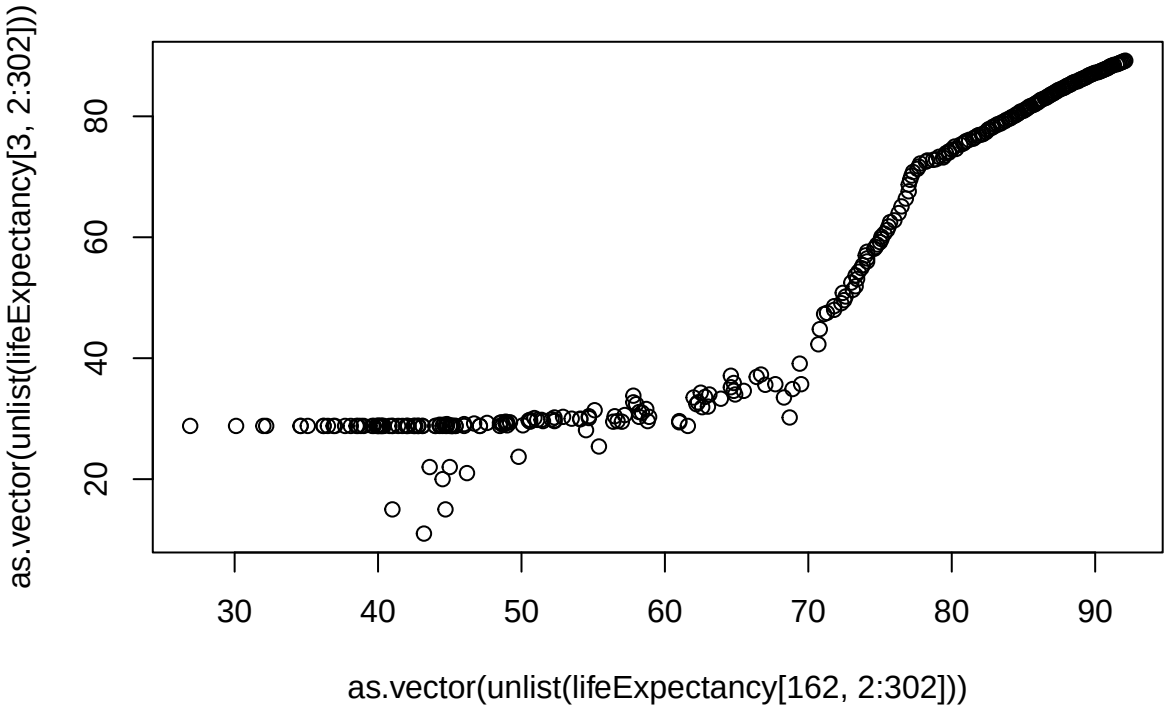
```r
plot(lifeExpectancy[,2],lifeExpectancy[,101])
```




```r
plot(lifeExpectancy[,2],lifeExpectancy[,3],xlab="1800", ylab="1801")
```




```r
plot(as.vector(unlist(lifeExpectancy[162,2:302])),as.vector(unlist(lifeExpectancy[3,2:302])))
```

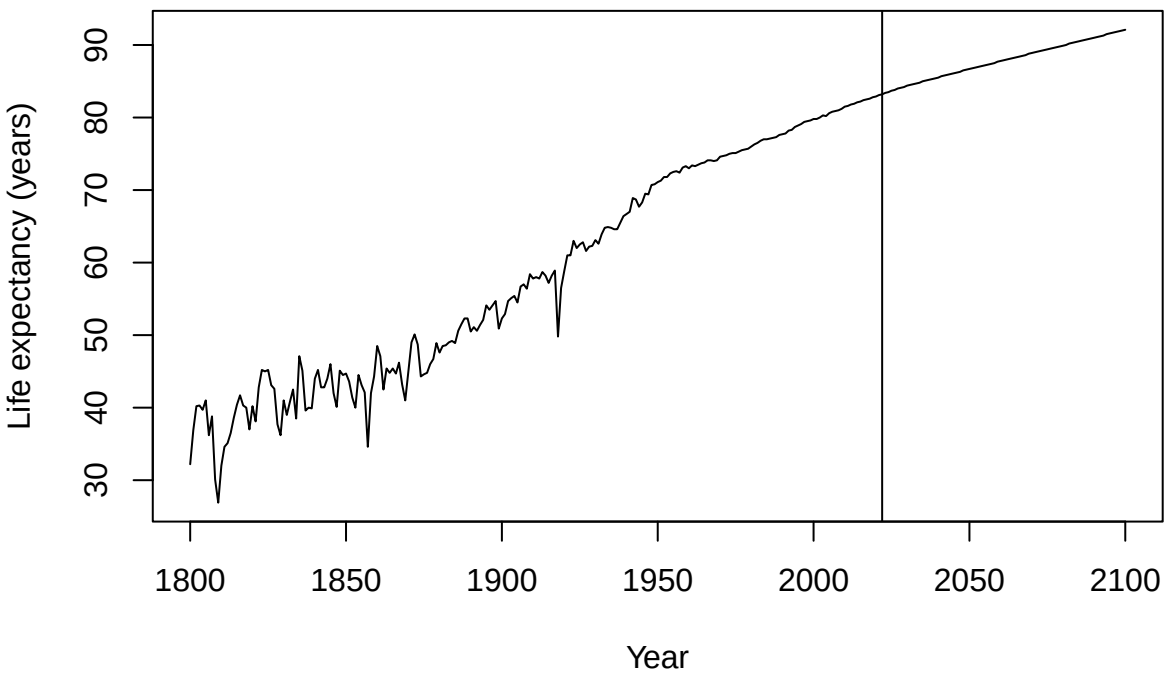



```
plot(as.vector(unlist(lifeExpectancy[162,2:302])),
     as.vector(unlist(colnames(lifeExpectancy)[2:302]))
     ,xlab="Life expectancy in years", ylab="Year")
abline(h=2022)
```

```
years <- as.numeric(colnames(lifeExpectancy)[2:302])
le <- as.numeric(lifeExpectancy[162, 2:302])
```

```r
plot(years, le,
     xlab = "Year",
     ylab = "Life expectancy (years)",
     type = "l")

abline(v = 2022)
```




```r
lifeExpectancy[lifeExpectancy$country=="Sweden",]
```

```
##     country 1800 1801 1802 1803 1804 1805 1806 1807 1808 1809 1810 1811 1812
## 162  Sweden 32.2 36.9 40.2 40.3 39.7   41 36.2 38.8 30.1 26.9   32 34.6 35.1
##     1813 1814 1815 1816 1817 1818 1819 1820 1821 1822 1823 1824 1825 1826 1827
## 162 36.5 38.6 40.4 41.7 40.3   40   37 40.2 38.1 42.8 45.2   45 45.2 43.1 42.6
##     1828 1829 1830 1831 1832 1833 1834 1835 1836 1837 1838 1839 1840 1841 1842
## 162 37.7 36.2   41   39 40.8 42.5 38.5 47.1 45.1 39.6   40 39.9   44 45.2 42.8
##     1843 1844 1845 1846 1847 1848 1849 1850 1851 1852 1853 1854 1855 1856 1857
## 162 42.8   44   46   42 40.1 45.1 44.5 44.7 43.6 41.4   40 44.5 43.1 42.1 34.6
##     1858 1859 1860 1861 1862 1863 1864 1865 1866 1867 1868 1869 1870 1871 1872
## 162   42 44.3 48.5 47.1 42.5 45.4 44.8 45.4 44.7 46.2 43.2   41   45   49 50.1
##     1873 1874 1875 1876 1877 1878 1879 1880 1881 1882 1883 1884 1885 1886 1887
## 162 48.7 44.3 44.6 44.8   46 46.7 48.9 47.6 48.5 48.6   49 49.2 48.9 50.6 51.5
##     1888 1889 1890 1891 1892 1893 1894 1895 1896 1897 1898 1899 1900 1901 1902
## 162 52.3 52.3 50.5 51.1 50.6 51.4 52.1 54.1 53.5 54.1 54.7 50.9 52.3 52.9 54.7
##     1903 1904 1905 1906 1907 1908 1909 1910 1911 1912 1913 1914 1915 1916 1917
## 162 55.1 55.4 54.5 56.7   57 56.4 58.4 57.8   58 57.8 58.7 58.2 57.2 58.2 58.9
##     1918 1919 1920 1921 1922 1923 1924 1925 1926 1927 1928 1929 1930 1931 1932
## 162 49.8 56.5 58.8   61   61   63   62 62.5 62.8 61.6 62.2 62.3 63.1 62.6 63.9
##     1933 1934 1935 1936 1937 1938 1939 1940 1941 1942 1943 1944 1945 1946 1947
## 162 64.8 64.9 64.8 64.6 64.6 65.5 66.4 66.7   67 68.9 68.7 67.7 68.3 69.5 69.4
##     1948 1949 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961 1962
## 162 70.7 70.8 71.1 71.3 71.8 71.8 72.3 72.5 72.6 72.4 73.1 73.3   73 73.4 73.3
```

```
##     1963 1964 1965 1966 1967 1968 1969 1970 1971 1972 1973 1974 1975 1976 1977
## 162 73.5 73.7 73.8 74.1 74.1   74 74.1 74.6 74.7 74.8   75 75.1 75.1 75.3 75.5
##     1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992
## 162 75.6 75.7   76 76.3 76.5 76.8   77   77 77.1 77.2 77.3 77.6 77.7 77.8 78.2
##     1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007
## 162 78.3 78.7 78.9 79.1 79.4 79.5 79.6 79.8 79.8   80 80.3 80.2 80.6 80.8 80.9
##     2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022
## 162   81 81.2 81.5 81.6 81.8 81.9 82.1 82.2 82.4 82.5 82.6 82.8 82.9 83.1 83.2
##     2023 2024 2025 2026 2027 2028 2029 2030 2031 2032 2033 2034 2035 2036 2037
## 162 83.4 83.5 83.7 83.8   84 84.1 84.2 84.4 84.5 84.6 84.7 84.8   85 85.1 85.2
##     2038 2039 2040 2041 2042 2043 2044 2045 2046 2047 2048 2049 2050 2051 2052
## 162 85.3 85.4 85.5 85.7 85.8 85.9   86 86.1 86.2 86.3 86.5 86.6 86.7 86.8 86.9
##     2053 2054 2055 2056 2057 2058 2059 2060 2061 2062 2063 2064 2065 2066 2067
## 162   87 87.1 87.2 87.3 87.4 87.5 87.7 87.8 87.9   88 88.1 88.2 88.3 88.4 88.5
##     2068 2069 2070 2071 2072 2073 2074 2075 2076 2077 2078 2079 2080 2081 2082
## 162 88.6 88.8 88.9   89 89.1 89.2 89.3 89.4 89.5 89.6 89.7 89.8 89.9   90 90.2
##     2083 2084 2085 2086 2087 2088 2089 2090 2091 2092 2093 2094 2095 2096 2097
## 162 90.3 90.4 90.5 90.6 90.7 90.8 90.9   91 91.1 91.2 91.3 91.5 91.6 91.7 91.8
##     2098 2099 2100
## 162 91.9   92 92.1
```

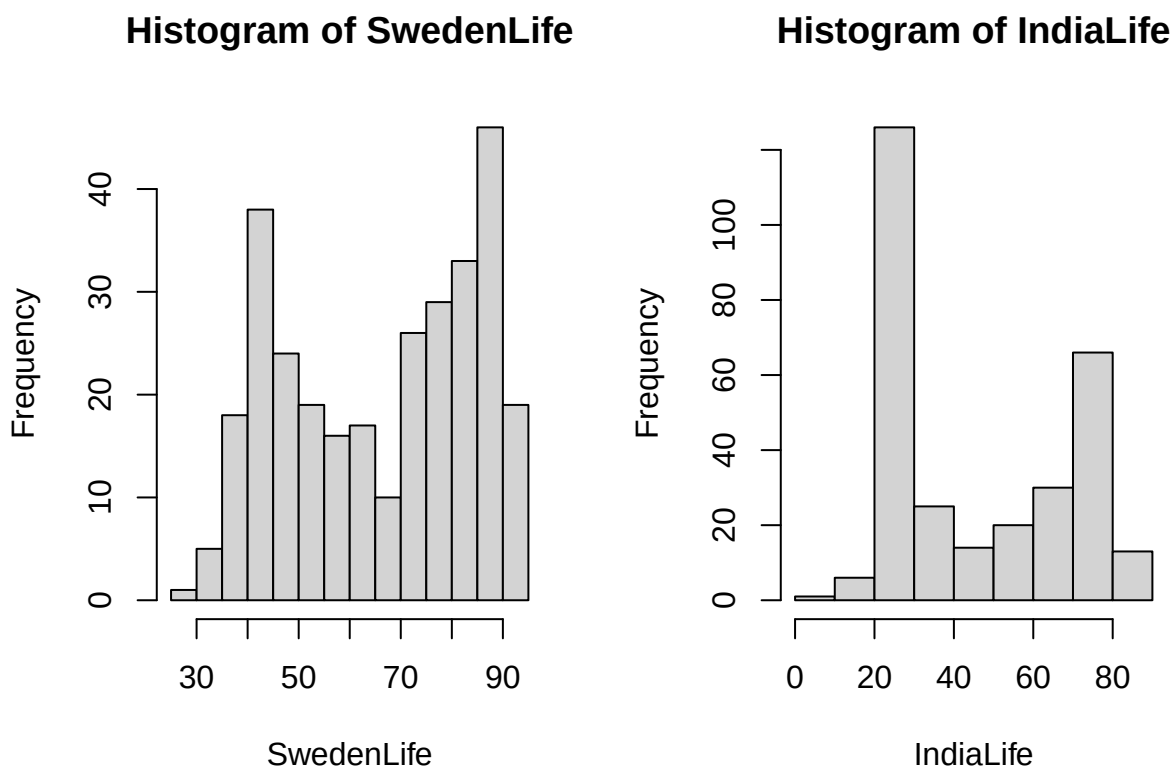
```r
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="Sweden",])))->SwedenLife
```

```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```r
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="India",])))->IndiaLife
```

```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```r
par(mfrow=c(1,2))
hist(SwedenLife)
hist(IndiaLife)
```




```
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="Bangladesh",])))->BanglaLife
```

```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="Australia",])))->AussieLife
```

```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="China",])))->ChinaLife
```

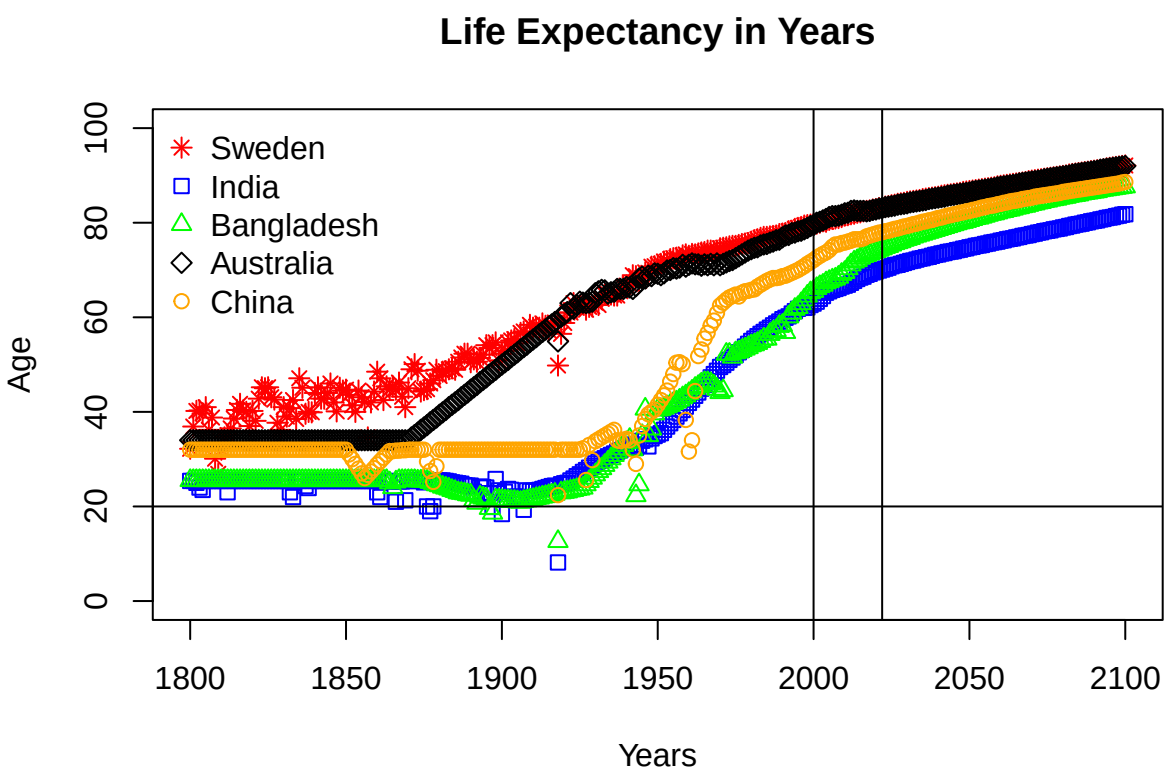
```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

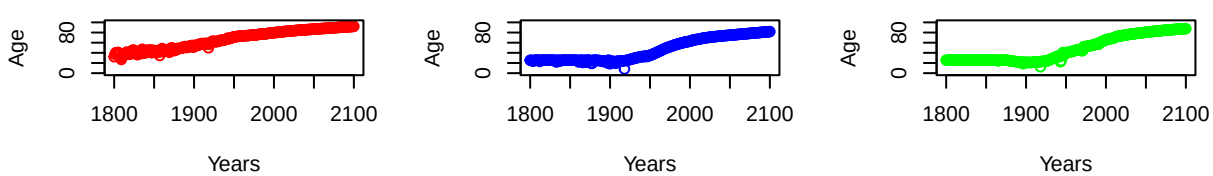
```
plot(c(1800:2100),SwedenLife,col="red",main="Life Expectancy in Years",pch=8,ylim=c(0,100),xlab="Years"
points(c(1800:2100),IndiaLife,col="blue", pch = 0)
points(c(1800:2100),BanglaLife,col="green",  pch = 2)
points(c(1800:2100),AussieLife,col="black", pch = 5)
points(c(1800:2100),ChinaLife,col="orange", pch =1)

legend("topleft",
       legend = c("Sweden", "India", "Bangladesh", "Australia", "China"),
       col    = c("red", "blue", "green", "black", "orange"),
       pch    = c(8, 0 , 2, 5, 1),
       bty    = "n")
```

```r
abline(v=2000)
abline(v=2022)

abline(h=20)
```




```r
par(mfrow=c(3,3))
plot(c(1800:2100),SwedenLife,col="red",ylim=c(0,100),xlab="Years",ylab="Age")
plot(c(1800:2100),IndiaLife,col="blue",ylim=c(0,100),xlab="Years",ylab="Age")
plot(c(1800:2100),BanglaLife,col="green",ylim=c(0,100),xlab="Years",ylab="Age")
```